CSE 270 Sports Analytics

Measuring individual performance in team sports

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NBA Individual Power Ratings

NBA Individual Power Ratings

Libraries used

```
library(SportsAnalytics270)
library(dplyr)
library(ggplot2)
library(VGAM)
library(zoo)
library(tidyr)
# remotes::install_github("Habet/CSE270")
```

NBA Individual Power Ratings

2 main stages of the game.

- Attacking
- Defending

https://www.youtube.com/watch?v=wYjp2zoqQrs

NBA Individual Power Ratings

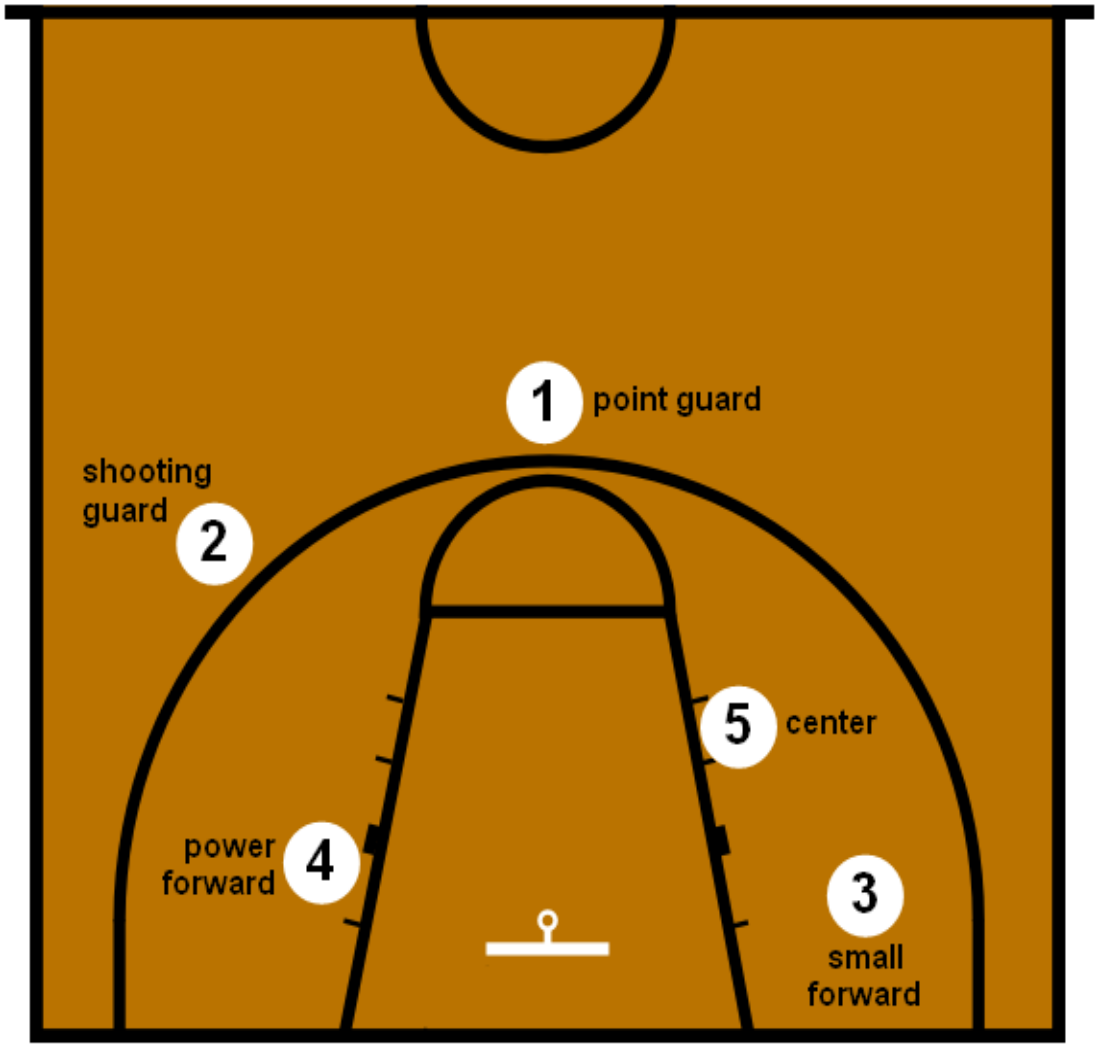
5 main positions on the field.

- Center - On offense, the center tries to score on close shots and rebound. But on defence, the center tries to block opponents' shots and rebound their misses.
- Power Forward - The power forward does many of the things a center does, playing near the basket while rebounding and defending taller players. But power forwards also take longer shots than centers.
- Small Forward - The small forward plays against small and large players. They roam all over on the court. Small forwards can score from long shots and close ones.
- Point Guard - The point guard runs the offense.The point guard defends the opponent's point guard and tries to steal the ball.
- Shooting Guard - The shooting guard can make shots from long distance and also is a good dribbler.

Let's watch a short video,

https://www.youtube.com/watch?v=4_4CymXARWQ

NBA Individual Power Ratings



NBA Individual Power Ratings

Defining attributes for each position.

- Center - The center is the tallest player on each team, playing near the basket.
- Power Forward - The power forward is very close to center player, but in this position they should also have good shooting attributes.
- Small Forward - The small forward should have good shooting efficiency in close and far distances.
- Point Guard - The point guard should have good defending characteristics.
- Shooting Guard - The shooting guard should have good dribbling attributes and shooting accuracy in long ranges.

What measures other than the height and shooting can be used to position the players on the field ?

NBA Individual Power Ratings

In game statistics

- STEALS
- TURNOVERS - Turnover occurs when a team loses possession of the ball to the opposing team before a player takes a shot at their team's basket.
  https://www.youtube.com/shorts/I4MDXOWrUmk
- REBOUNDS
  1. OFFENSIVE REBOUNDS - An offensive rebound occurs when offensive players regain possession of the ball after a missed shot.
  2. DEFENSIVE REBOUNDS - Defensive rebounds occur when defensive players gain possession of the ball after an offensive player misses the shot.
- BLOCKS - Block occurs when a defensive player legally deflects a field goal attempt from an offensive player to prevent a score.
- FOULS
  1. PERSONAL FOULS
  2. FLAGRANT FOULS
  3. TECHINCAL FOULS

NBA Individual Power Ratings

Dataset

`nba_players` runs through the 2024-25 season. `Year` is the season ending year, so 2025 is that season.

```
str(nba_players)
```

```
## 'data.frame':    16955 obs. of  53 variables:
##  $ X     : int  13600 13712 13713 13711 13932 13656 13578 13490 13615 13627 ...
##  $ Year  : int  1998 1998 1998 1998 1998 1998 1998 1998 1998 1998 ...
##  $ Player: chr  "A.C. Green" "Aaron McKie" "Aaron McKie" "Aaron McKie" ...
##  $ Pos   : chr  "PF" "SF" "SG" "SG-SF" ...
##  $ Age   : num  34 25 25 25 26 27 22 31 25 26 ...
##  $ Tm    : chr  "DAL" "DET" "PHI" "TOT" ...
##  $ G     : num  82 24 57 81 65 80 55 1 69 82 ...
##  $ GS    : num  68 1 31 32 9 75 1 0 33 82 ...
##  $ MP    : num  2649 472 1341 1813 757 ...
##  $ PER   : num  12.8 10.1 8.1 8.6 13.4 14.5 9.2 0 18.4 15.1 ...
##  $ TS.   : num  0.496 0.478 0.383 0.41 0.575 0.605 0.418 NA 0.534 0.53 ...
##  $ X3PAr : num  0.007 0.163 0.166 0.165 0.005 0 0.006 NA 0.008 0.167 ...
##  $ FTr   : num  0.303 0.221 0.116 0.144 0.386 0.472 0.365 NA 0.515 0.262 ...
##  $ ORB.  : num  9 6 2.9 3.7 7.7 11.6 11.5 0 12.2 1.8 ...
##  $ DRB.  : num  19.3 11.2 11.1 11.2 15.4 14.8 18.6 0 13.8 9.8 ...
##  $ TRB.  : num  14 8.6 7 7.4 11.6 13.3 15 0 13 5.8 ...
##  $ AST.  : num  7.3 12.9 15.8 15 3 6.7 3.5 0 6.7 14.3 ...
##  $ STL.  : num  1.6 2.7 3.1 3 1.3 1.4 1 0 1.1 1.2 ...
##  $ BLK.  : num  0.7 0.2 0.7 0.5 3.8 0.9 5.4 0 1.3 0.7 ...
##  $ TOV.  : num  10.1 17.4 15.2 15.8 16.3 12.2 20.2 NA 10.6 12.3 ...
##  $ USG.  : num  11.6 13.6 11.5 12 18.8 13.2 16.4 0 23.9 26.7 ...
##  $ blanl : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ OWS   : num  2.8 0.1 -0.8 -0.7 0.7 4.8 -1 0 4.2 2.5 ...
##  $ DWS   : num  2.4 0.6 1.4 2 1 1.7 0.9 0 1.8 3.5 ...
##  $ WS    : num  5.2 0.7 0.7 1.4 1.7 6.5 -0.1 0 6 6.1 ...
##  $ WS.48 : num  0.095 0.069 0.024 0.036 0.105 0.153 -0.009 -0.022 0.144 0.102 ...
##  $ blank2: num  NA NA NA NA NA NA NA NA NA NA ...
##  $ OBPM  : num  -1.7 -1.7 -3.2 -2.8 -3.4 1.9 -6.9 -7.3 0.9 0.1 ...
```

NBA Individual Power Ratings

Let's select the performance metrics.

```
nba_players <- nba_players %>%
  select(Year, Player, Pos,
         G, GS, MP, STL, TOV, ORB, DRB, PF, BLK, PTS, AST,
         FT, FTA, FG, FGA, X3P, X3PA, X2P, X2PA, ORB., DRB.,
         TRB., AST., STL., BLK., TOV., FG., X3P., FT.,) %>%
  arrange(Player, Year)
head(nba_players, 3)
```

```
##   Year     Player Pos  G GS   MP STL TOV ORB DRB  PF BLK PTS AST  FT FTA  FG
## 1 1998 A.C. Green  PF 82 68 2649  78  68 219 449 157  27 600 123 116 162 242
## 2 1999 A.C. Green  PF 50 35  924  28  19  82 146  69   8 246  25  30  52 108
## 3 2000 A.C. Green  PF 82 82 1929  53  53 160 326 127  18 413  80  66  95 173
##   FGA X3P X3PA X2P X2PA ORB. DRB. TRB. AST. STL. BLK. TOV.   FG. X3P.   FT.
## 1 534   0    4 242  530  9.0 19.3 14.0  7.3  1.6  0.7 10.1 0.453 0.00 0.716
## 2 256   0    8 108  248  9.9 17.8 13.9  4.5  1.6  0.6  6.4 0.422 0.00 0.577
## 3 387   1    4 172  383  9.0 17.9 13.5  5.9  1.4  0.7 11.0 0.447 0.25 0.695
```

NBA Individual Power Ratings

Let's try to evaluate the player's performance based on the amount of time they spent on the field.
We will use the players' historical data to measure their performance for each Year, by taking 5 preceding Years only in order to avoid one Year stars. We will use success/failure approach by dividing the metrics of the players into successful and non-succesfull actions and then compare their success rate with their playing Minutes.

NBA Individual Power Ratings

Let's start with defending attributes.

Successful actions, when defending

- STEALS
- BLOCKS
- DEFENSIVE REBOUNDS

Non Successful actions, when defending

- PERSONAL FOULS
- TURNOVER

NBA Individual Power Ratings

Let's calculate the difference between the cumulative success actions and cumulative failure actions.

```
nba_players_defence <- nba_players %>%
  arrange(Player, Year) %>%
  filter(Pos %in% c('SF','PG')) %>%
  group_by(Player) %>%
  mutate(STL_TOV = STL - TOV,
         STL_PF = STL - PF,
         DRB_TOV = DRB - TOV,
         BLK_TOV = BLK - TOV,
         BLK_PF = BLK - PF) %>%
  drop_na(STL_TOV, STL_PF, DRB_TOV, BLK_TOV, BLK_PF, MP) %>%
  ungroup()
```

NBA Individual Power Ratings

Let's check if the metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_defence$MP))
}

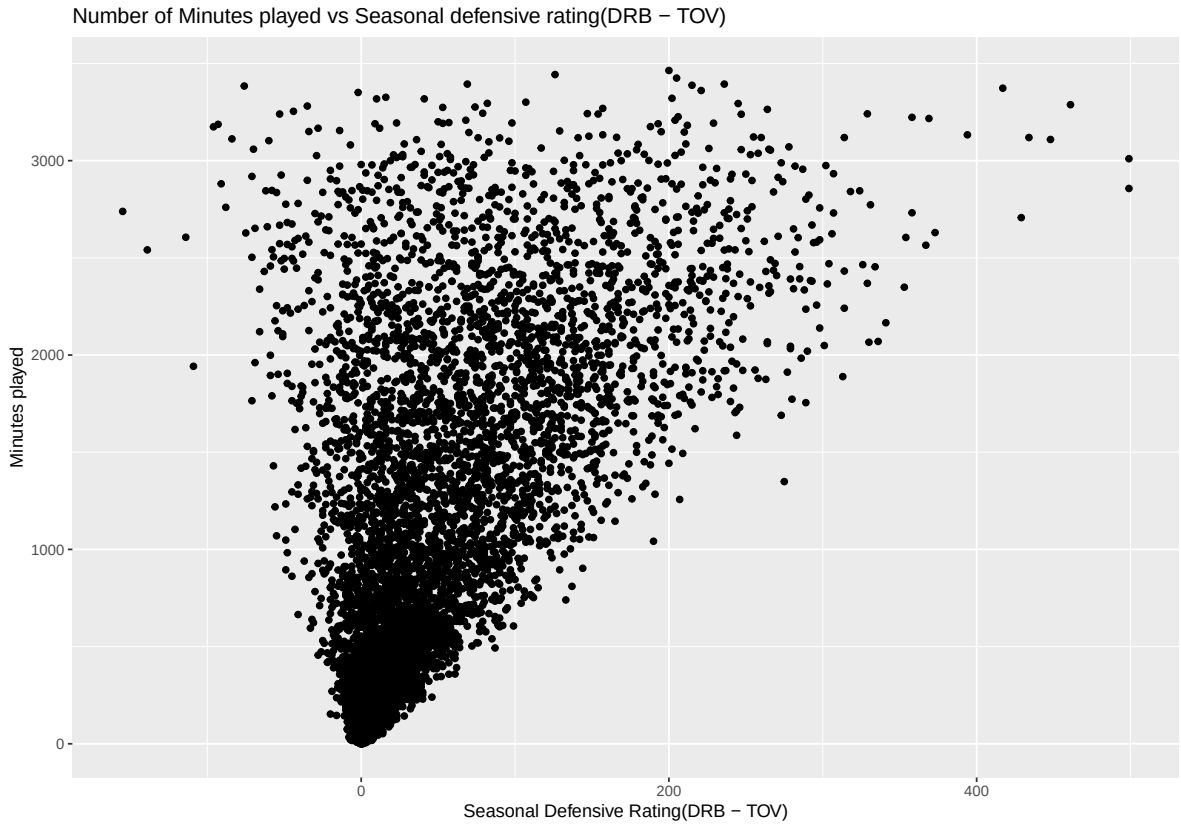
nba_players_defence %>%
  select(STL_TOV, STL_PF, DRB_TOV, BLK_TOV, BLK_PF) %>%
  apply(2,cor_with_MP)
```

```
##    STL_TOV     STL_PF    DRB_TOV    BLK_TOV     BLK_PF
## -0.6490088 -0.6950625  0.5825263 -0.8132951 -0.8773959
```

NBA Individual Power Ratings

The metric with the highest positive correlation rate is the Seasonal difference between the seasonal number of defensive rebounds made and number of turnovers made.

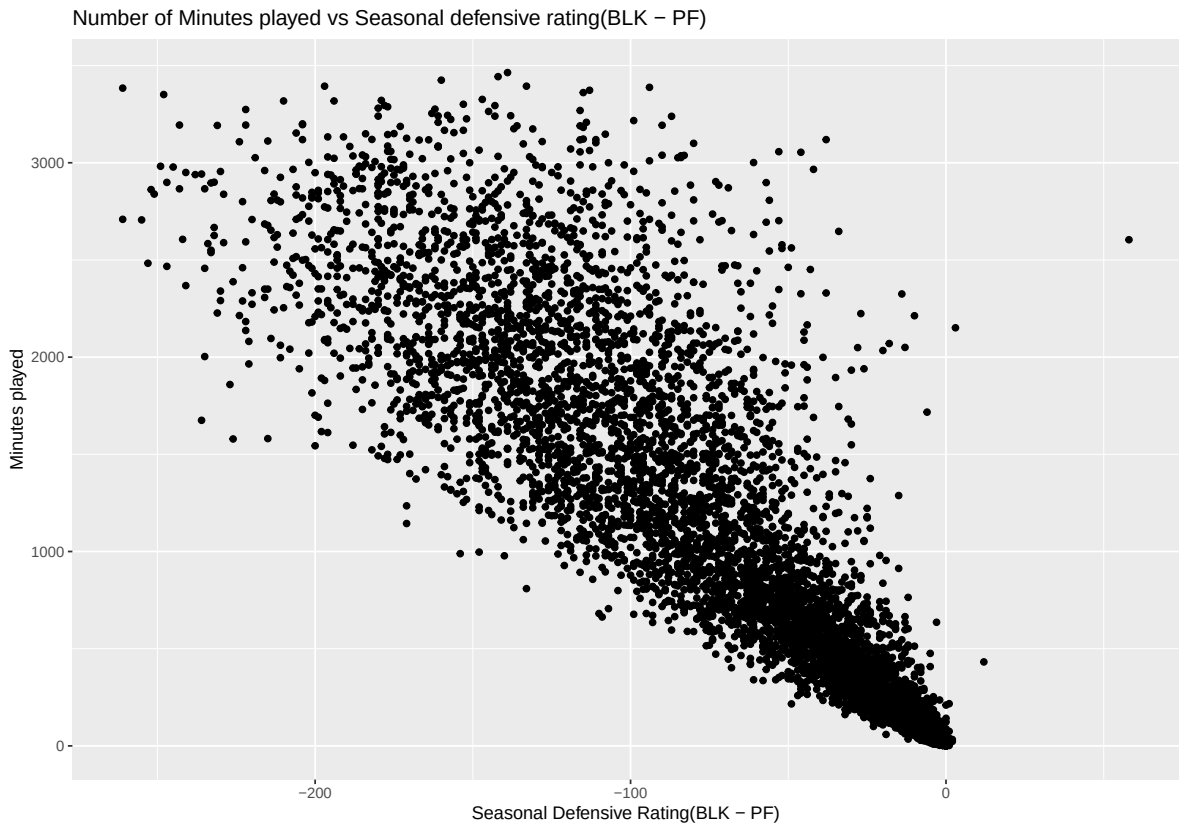
```
ggplot(nba_players_defence, aes(x = DRB_TOV, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(DRB - TOV)") +
  xlab("Seasonal Defensive Rating(DRB - TOV)") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

The metric with the highest negative correlation is blocks minus personal fouls.

```
ggplot(nba_players_defence, aes(x = BLK_PF, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(BLK - PF)") +
  xlab("Seasonal Defensive Rating(BLK - PF)") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

Let's fit a model with the defensive power ratings.

```
summary(lm(MP ~ (STL_TOV + STL_PF + DRB_TOV + BLK_PF ), data = nba_players_defence))
```

```
##
## Call:
## lm(formula = MP ~ (STL_TOV + STL_PF + DRB_TOV + BLK_PF), data = nba_players_defence)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2405.43  -111.51   -56.29    91.06  2330.39
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  74.40429    5.42980   13.70   <2e-16 ***
## STL_TOV      -6.75886    0.10654  -63.44   <2e-16 ***
## STL_PF        8.30940    0.17114   48.55   <2e-16 ***
## DRB_TOV       4.37224    0.05939   73.61   <2e-16 ***
## BLK_PF      -13.91125    0.12813 -108.57   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 273.5 on 6516 degrees of freedom
## Multiple R-squared:  0.9045, Adjusted R-squared:  0.9045
## F-statistic: 1.543e+04 on 4 and 6516 DF,  p-value: < 2.2e-16
```

NBA Individual Power Ratings

- Majority of the players make many fouls, but still play relatively often if they make more steals.
- Having more turnovers mostly causes less playing practice.
- Having more defensive rebounds and steals can improve player's playing practice.

NBA Individual Power Ratings

For the attacking attributes we will calculate the difference between the attempts made for successful actions and actual successful actions.

Successful actions, when attacking

- FIELD GOALS
- 3 POINT GOALS
- FREE THROWS
- OFFENSIVE REBOUNDS
- POINTS SCORED
- ASSISTS MADE

Non Successful actions, when attacking

- FIELD GOALS ATTEMPTS
- 3 POINT ATTEMPTS
- FREE THROW ATTEMPTS

NBA Individual Power Ratings

```
nba_players_offense <- nba_players %>%
  arrange(Player, Year)  %>%
  group_by(Player) %>%
  mutate(FGA_FG = FGA - FG,
         X3PA_X3P = X3PA - X3P,
         FTA_FT = FTA - FT) %>%
  drop_na(FGA_FG, X3PA_X3P, FTA_FT, AST, PTS, MP) %>%
  ungroup()
```

NBA Individual Power Ratings

Let's check if the Seasonal metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_offense$MP))
}

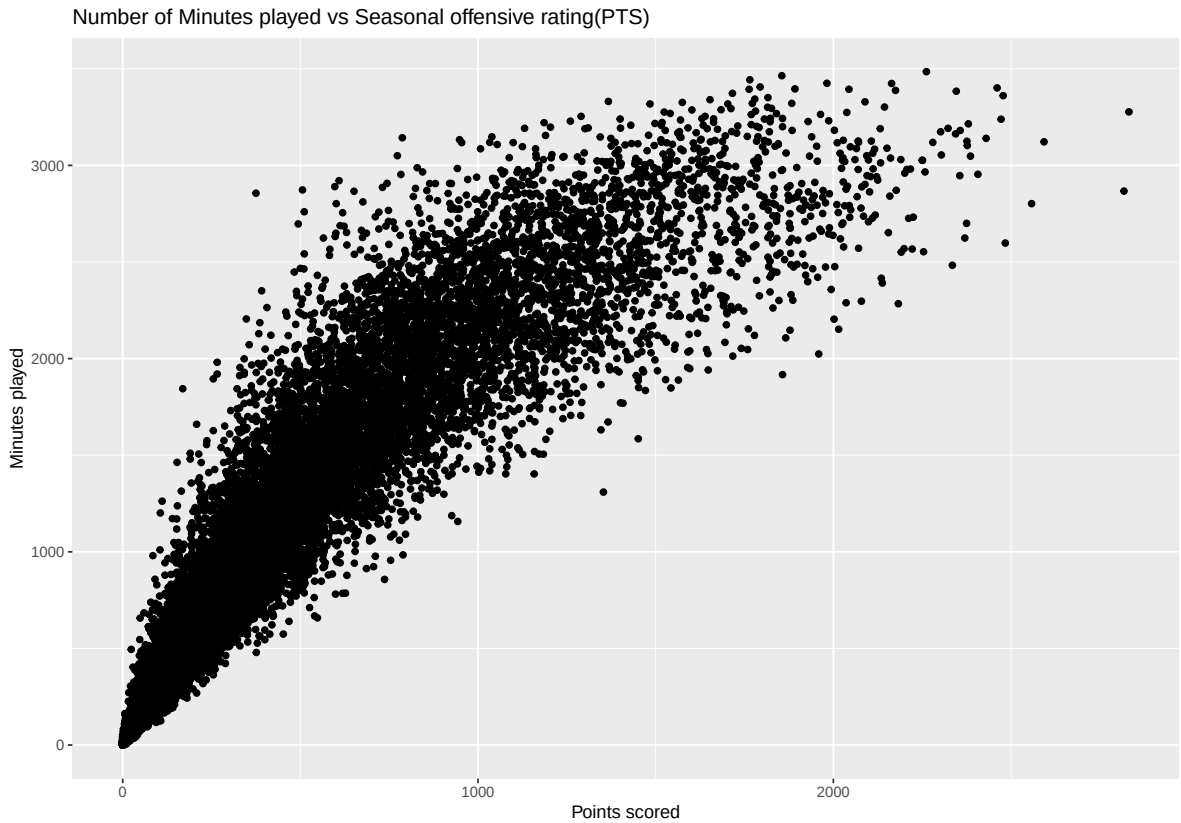
nba_players_offense %>%
  select(FGA_FG, X3PA_X3P, FTA_FT, AST, PTS) %>%
  apply(2,cor_with_MP)
```

```
##    FGA_FG  X3PA_X3P    FTA_FT       AST       PTS
## 0.9233937 0.6532211 0.7078765 0.7545190 0.9233347
```

NBA Individual Power Ratings

Missed field goals (FGA - FG) and points scored have the strongest correlations with minutes, both about 0.92. The plot shows points.

```
ggplot(nba_players_offense,aes(x = PTS, y = MP)) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal offensive rating(PTS)") +
  xlab("Points scored") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

```
summary(lm(MP ~ (PTS + AST + FGA_FG + X3PA_X3P + FTA_FT), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MP ~ (PTS + AST + FGA_FG + X3PA_X3P + FTA_FT), data = nba_players_offense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2307.83  -189.34   -67.36   156.27  1635.69
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 215.99824    3.40124  63.506  < 2e-16 ***
## PTS           0.47187    0.02557  18.453  < 2e-16 ***
## AST           0.68208    0.03050  22.365  < 2e-16 ***
## FGA_FG        2.44019    0.05819  41.932  < 2e-16 ***
## X3PA_X3P     -0.16328    0.05241  -3.115  0.00184 **
## FTA_FT        3.46688    0.13691  25.323  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 304.4 on 16941 degrees of freedom
## Multiple R-squared:  0.8754, Adjusted R-squared:  0.8754
## F-statistic: 2.38e+04 on 5 and 16941 DF,  p-value: < 2.2e-16
```

NBA Individual Power Ratings

- Scoring more points/assisting = playing more
- Attempting more to score a field goal or a field throw.

Plus/Minus

Plus/Minus

The concept

Plus-minus measures the point differential for the team when the player is on the court,
Point differential is the difference in the points scored for and the points scored against the team.
The NHL's Montreal Canadiens were the first team to track the plus - minus of its players, starting sometime in the 1950s. Other teams followed in the early 1960s, and the NHL started officially compiling the statistic for the 1967 - 68 season.

Plus/minus for NBA

```
colnames(nba12)
```

```
##  [1] "GameID"           "Date"             "AwayT"            "HomeT"
##  [5] "StartTime"        "EndTime"          "ElapsedTime"      "ElapsedSecs"
##  [9] "HomePlayer1ID"    "HomePlayer2ID"    "HomePlayer3ID"    "HomePlayer4ID"
## [13] "HomePlayer5ID"    "AwayPlayer1ID"    "AwayPlayer2ID"    "AwayPlayer3ID"
## [17] "AwayPlayer4ID"    "AwayPlayer5ID"    "HomePlayer1Name"  "HomePlayer2Name"
## [21] "HomePlayer3Name"  "HomePlayer4Name"  "HomePlayer5Name"  "AwayPlayer1Name"
## [25] "AwayPlayer2Name"  "AwayPlayer3Name"  "AwayPlayer4Name"  "AwayPlayer5Name"
## [29] "StartScoreHome"   "StartScoreAway"   "EndScoreHome"     "EndScoreAway"
## [33] "PointsScoredHome" "PointsScoredAway" "PossessionsHome"  "PossessionsAway"
```

Plus/minus for NBA

Subseting the data frame for Oklahoma City Thunders

```
oks_g <- grepl("OKC", nba12$GameID, fixed = TRUE)
okc<-nba12[oks_g == T,]
```

```
head(okc[,19:23], n=5)
```

```
##     HomePlayer1Name HomePlayer2Name HomePlayer3Name HomePlayer4Name
## 144       Westbrook       Sefolosha          Durant           Ibaka
## 145       Westbrook       Sefolosha          Durant           Ibaka
## 146       Westbrook          Harden          Durant           Ibaka
## 147       Westbrook          Harden          Durant           Ibaka
## 148       Westbrook          Harden          Durant           Ibaka
##     HomePlayer5Name
## 144         Perkins
## 145         Perkins
## 146         Perkins
## 147        Collison
## 148        Collison
```

Plus/minus for NBA

Create the line-ups variable

```
okc$line_h <-
  paste(okc$HomePlayer1Name, okc$HomePlayer2Name,
        okc$HomePlayer3Name, okc$HomePlayer4Name,
        okc$HomePlayer5Name, sep = ",")

okc$line_a <-
  paste(okc$AwayPlayer1Name, okc$AwayPlayer2Name,
        okc$AwayPlayer3Name,okc$AwayPlayer4Name,
        okc$AwayPlayer5Name,   sep = ",")

okc$line_h[1:5]
```

```
## [1] "Westbrook,Sefolosha,Durant,Ibaka,Perkins"
## [2] "Westbrook,Sefolosha,Durant,Ibaka,Perkins"
## [3] "Westbrook,Harden,Durant,Ibaka,Perkins"
## [4] "Westbrook,Harden,Durant,Ibaka,Collison"
## [5] "Westbrook,Harden,Durant,Ibaka,Collison"
```

Plus/minus for NBA

Calculating plus/minus variable for time stunts

Away team

```
# Home team plus minus
okc$Home_PM <- okc$PointsScoredHome - okc$PointsScoredAway
# Away team plus minus
okc$Away_PM <- -okc$Home_PM
```

Check if Durant is playing

```
# is Durant playing at home?
okc$D_H <- grepl("Durant", okc$line_h, fixed = TRUE)
# is Durant playing away?
okc$D_A <- grepl("Durant", okc$line_a, fixed = TRUE)
```

Plus/minus for NBA

Get the plus minus for Durant for the whole season

```
sum(okc$Home_PM[okc$D_H == T]) + sum(okc$Away_PM[okc$D_A == T])
```

```
## [1] 324
```

This will be the plus/minus for the whole season

Plus/minus for NBA

For each game

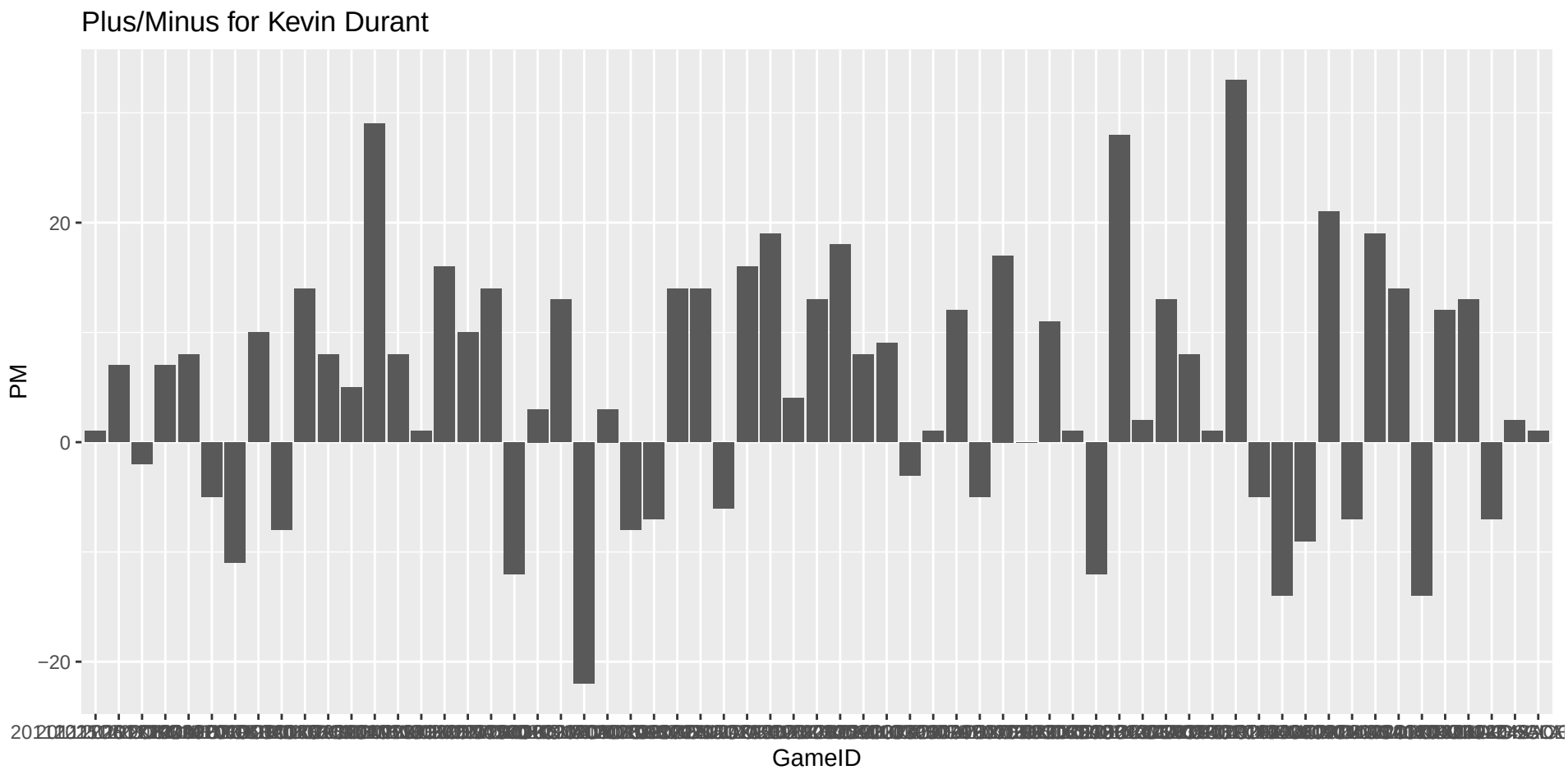
```
durant_pm <- okc %>%
  mutate(PM=ifelse(D_H==T,
                   Home_PM,
                   ifelse(D_A==T, Away_PM,0))) %>%
  group_by(GameID) %>%
  summarise(PM=sum(PM))
```

```
head(durant_pm, n=5)
```

```
## # A tibble: 5 x 2
##   GameID             PM
##   <fct>           <dbl>
## 1 20111225ORLOKC      1
## 2 20111226OKCMIN      7
## 3 20111228OKCMEM     -2
## 4 20111229DALOKC      7
## 5 20111231PHXOKC      8
```

Plus/minus for NBA

```
ggplot(durant_pm, aes(x=GameID, y=PM))+
  geom_bar(stat="identity")+
  ggtitle("Plus/Minus for Kevin Durant")
```



On court and off court +/- and Roland's rating

The on court +/- number represents the team's net points with the player on the floor per 48 minutes, while the off court number is the team's net with the player off the floor per 48 minutes.

The Roland Rating is the difference between the two, with a positive number indicating the team has played better with the player than without.

Net Plus/minus for NBA

Create a variable that shows OKC plus/minus, depending on whether it is playing at home or away

```
okc <- okc %>%
# makes PM for OKC
mutate(OKC_PM = ifelse(AwayT == "OKC", Away_PM,
                       ifelse(HomeT == "OKC", Home_PM, 0)),
       D_PM = ifelse(D_H == T | D_A == T, T, F)
       )
```

Net Plus/minus for NBA

For each game, D_PM has a value of FALSE if Durant has not played, and TRUE if He has. PM is showing the plus/minus for that game

```
durant_pm <- okc %>%
group_by(GameID, D_PM) %>%
summarise(PM = sum(OKC_PM))
head(durant_pm)
```

```
## # A tibble: 6 x 3
## # Groups:   GameID [3]
##   GameID          D_PM     PM
##   <fct>           <lgl> <dbl>
## 1 20111225ORLOKC FALSE     7
## 2 20111225ORLOKC TRUE      1
## 3 20111226OKCMIN FALSE    -3
## 4 20111226OKCMIN TRUE      7
## 5 20111228OKCMEM FALSE     5
## 6 20111228OKCMEM TRUE     -2
```

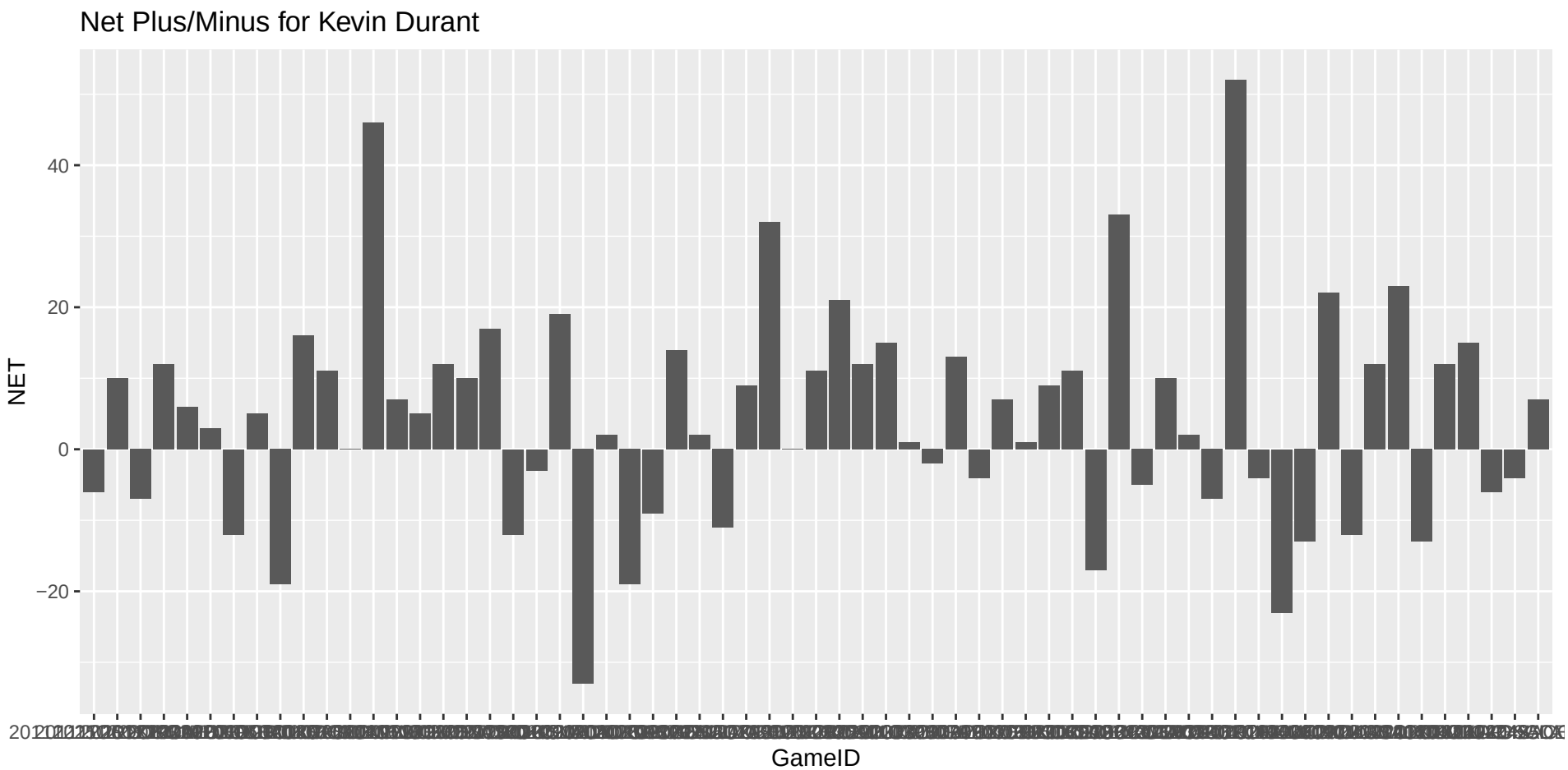
Net Plus/minus for NBA

```
durant_pm <- okc %>%
group_by(GameID, D_PM) %>%
summarise(PM = sum(OKC_PM)) %>%
group_by(GameID) %>%
mutate(NET = PM - lag(PM, 1)) %>%
filter(NET != "NA") %>%
select(GameID, NET)
head(durant_pm)
```

```
## # A tibble: 6 x 2
## # Groups:   GameID [6]
##   GameID           NET
##   <fct>          <dbl>
## 1 20111225ORLOKC    -6
## 2 20111226OKCMIN    10
## 3 20111228OKCMEM    -7
## 4 20111229DALOKC    12
## 5 20111231PHXOKC     6
## 6 20120102OKCDAL     3
```

Plus/minus for NBA

```
ggplot(durant_pm, aes(x=GameID, y=NET))+
  geom_bar(stat="identity")+
  ggtitle("Net Plus/Minus for Kevin Durant")
```



Adjusted plus/minus

Adjusted plus/minus

The plus/minus rating is adjusted by the players who are on the court

$$MARGIN = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... \beta_k X_k + \epsilon$$

where

$X_k = 1$, if player k is playing at home,
$X_k = -1$, if player k is playing away,
$X_k = 0$, if player k is not playing

Adjusted plus/minus

According to Dan Rosenbaum's approach, β coefficients will show the adjusted plus/minus of the players, or

β_k measures the difference between player k and the reference players, holding the other k players constant

Adjusted plus/minus

Data preparation/ this will take at least 30-40 minutes on the average machine

Data prep for adjusted plus/minus

```
# Get home and away teams
nba3 <- nba12[, c("HomePlayer1ID", "HomePlayer2ID",
                  "HomePlayer3ID", "HomePlayer4ID", "HomePlayer5ID")]

nba4 <- nba12[, c("AwayPlayer1ID", "AwayPlayer2ID",
                  "AwayPlayer3ID", "AwayPlayer4ID","AwayPlayer5ID")]

# MAke ID as character
nba3 <- as.data.frame(sapply(nba3, as.character))
nba4 <- as.data.frame(sapply(nba4, as.character))

#create dummy variables
nba3 <- as.data.frame(model.matrix(~. - 1, nba3))
nba4 <- as.data.frame(model.matrix(~. - 1, nba4))
# for away team
nba4[nba4==1]<- -1
# change colnames to be more readable
colnames(nba3) <- gsub("HomePlayer[123456789]ID", "", colnames(nba3))
colnames(nba4) <- gsub("AwayPlayer[123456789]ID", "", colnames(nba4))

# nba3 remove double entries
nba3 <- t(nba3)     # transpose the data.frame
nba3 <- by(nba3, INDICES=row.names(nba3), FUN=colSums) # collapse the rows with the same name
nba3 <- as.data.frame(do.call(cbind,aggr))
nba4 <- t(nba4)     # transpose the data.frame
nba4 <- by(nba4, INDICES=row.names(nba4), FUN=colSums) # collapse the rows with the same name
nba4 <- as.data.frame(do.call(cbind,nba4))

#combine home and away players and double columns
nba5<-data.frame(nba3,nba4)
nba5 <- t(nba5)     # transpose the data.frame
nba5 <- by(nba5, INDICES=row.names(nba5), FUN=colSums) # collapse the rows with the same name
nba5 <- as.data.frame(do.call(cbind,nba5))
```

Adjusted plus/minus

```
data(nba5, package = "SportsAnalytics270")
```

```
dim(nba5)
```

```
## [1] 25300   474
```

Adjusted plus/minus

Coefficients are named by player ID, we need to match with player names

Add plus/minus as dependent variable

```
nba5<-data.frame(PM=nba12$PointsScoredHome-nba12$PointsScoredAway, nba5)
```

Run the regression

```
model<-lm(PM~., data=nba5)
coeff<-sort(coefficients(model), decreasing=T)
coeff[1:10]
```

```
##     X1027      X453      X936      X678      X900      X158      X904      X781
## 2.1176032 1.8681832 1.5567575 1.0007997 0.9188331 0.8786333 0.8486712 0.8390129
##      X919      X964
## 0.8211618 0.8188564
```

Remove X from the begining of the coeff name and match with player info

```
name1<-names(coeff)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
```

Adjusted plus/minus

Corresponding dataframe with player level information

```
data(nba12p, package = "SportsAnalytics270")
```

```
head(nba12p[,1:3],5)
```

```
##   PlayerID   PlayerTrueName PlayerTeams
## 1        3   Dampier, Erick         ATL
## 2       17   Diop, DeSagana         CHA
## 3       35    Boykins, Earl         HOU
## 4       36  Najera, Eduardo         CHA
## 5       42 Elson, Francisco         PHI
```

Adjusted plus/minus

```
nba12p$PlayerID<-as.character(nba12p$PlayerID)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
name2[1:5,]
```

```
##   name1    PlayerTrueName
## 1  1027     Benson, Keith
## 2   453        Diogu, Ike
## 3   936   N'Diaye, Hamady
## 4   678    Williams, Sean
## 5   900 Whiteside, Hassan
```

rename coefficients by names

```
names(coeff)<-name2$PlayerTrueName
```

Adjusted plus/minus

Do you know this players?

Lets look at top ten players

```
coeff[1:10]
```

```
##     Benson, Keith         Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032          1.8681832          1.5567575          1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331          0.8786333          0.8486712          0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618          0.8188564
```

Benson, Keith - https://goo.gl/YxMsJ9

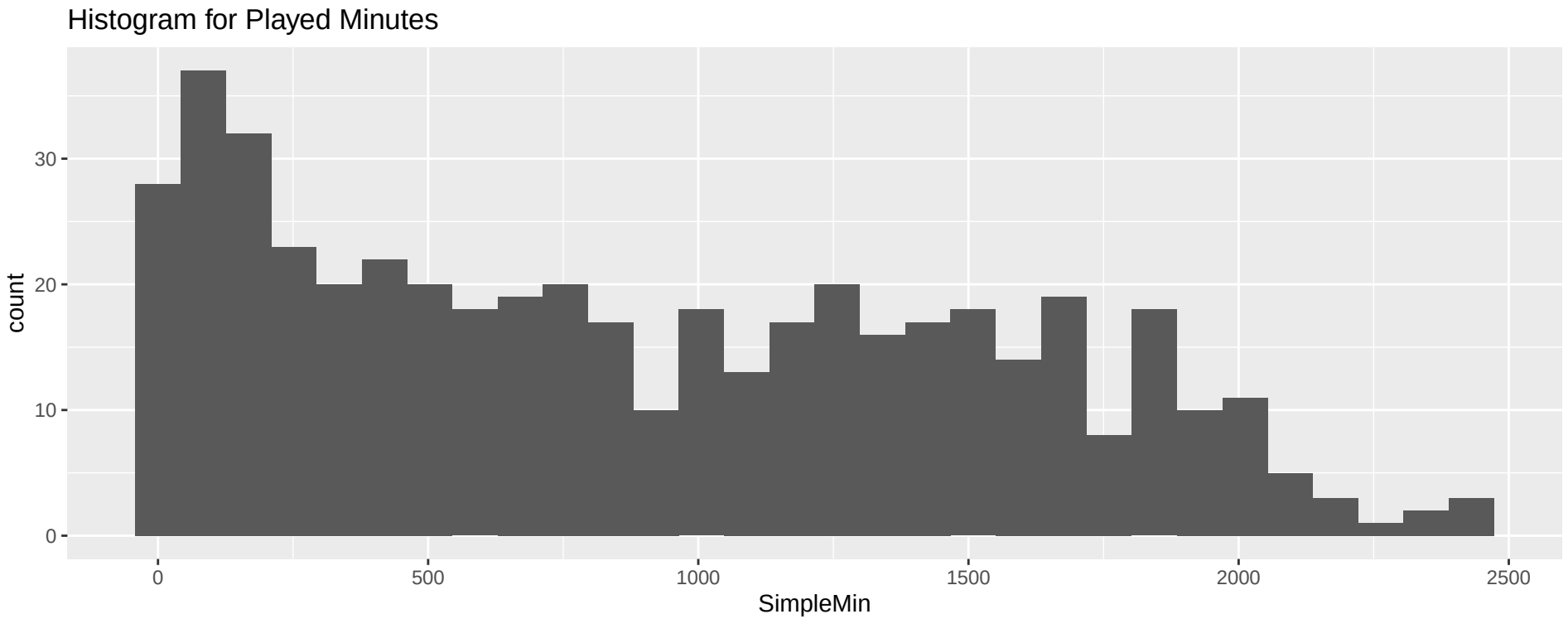
Adjusted plus/minus

Leave only players that has played more than 250 minutes (By Rosenbaum)

```
summary(nba12p$SimpleMin)
```

```
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0   292.1   812.5   892.0  1416.6  2430.9
```

```
ggplot(data=nba12p, aes(SimpleMin))+geom_histogram()+
  ggtitle("Histogram for Played Minutes")
```



Adjusted plus/minus

Getting players with more than 250m of play

```
more250<-nba12p$PlayerID[nba12p$SimpleMin>250]
length(more250)
```

```
## [1] 372
```

```
more250[1:10]
```

```
##  [1] "17"  "51"  "109" "132" "144" "211" "459" "484" "514" "550"
```

```
# Subset the dataframe by more250
colnames(nba5) <- gsub("X", "", colnames(nba5))
nba6<-nba5[,colnames(nba5) %in% more250]
nba6<-data.frame(PM=nba5$PM, nba6)
# Make the model
model2<-lm(PM~., data=nba6)
# get the coefficients
coeff2<-sort(coefficients(model2), decreasing=T)
```

Adjusted plus/minus

```
coeff2[1:10]
```

```
##      X158      X113      X807       X70      X144      X904      X821      X791
## 0.8779187 0.7791911 0.7605357 0.6760659 0.6705972 0.6580038 0.6435656 0.6088434
##      X938        X5
## 0.6001202 0.5689375
```

Adjusted plus/minus

Get the adjust plus/minus from coefficients

```
name1<-names(coeff2)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
name2$rank <- rownames(name2)
rownames(name2) <- c()
name2 <- name2 %>%
  tidyr::drop_na()
names(coeff2)<-name2$PlayerTrueName
```

Adjusted plus/minus

```
coeff2[1:10]
```

```
## Nelson, Jameer   Duhon, Chris Hudson, Lester    Paul, Chris   Battie, Tony
##      0.8779187      0.7791911      0.7605357      0.6760659      0.6705972
##  Bledsoe, Eric Curry, Stephen   Lopez, Robin     Wall, John Nowitzki, Dirk
##      0.6580038      0.6435656      0.6088434      0.6001202      0.5689375
```

Nelson Jameer https://goo.gl/eNj7sq

Adjusted plus/minus

Where is Durant ?

```
name2[name2$PlayerTrueName=="Durant, Kevin",]
```

```
##     name1 PlayerTrueName rank
## 224   714  Durant, Kevin  225
```

Where is Lebron ?

```
name2[name2$PlayerTrueName=="James, LeBron",]
```

```
##    name1 PlayerTrueName rank
## 15   260  James, LeBron   15
```

Adjusted plus/minus

- Modifications
  - Take only players that played more then 250 minutes during one season
  - As a depended variable, take margin of Plus/minus per 100 possessions

$$\text{Margin} = \frac{\text{Home Points}}{\text{Home Possession}} - \frac{\text{Away Points}}{\text{Away Possesssion}}$$

Adjusted plus/minus

```
str(nba12)
```

```
## 'data.frame':    25300 obs. of  36 variables:
##  $ GameID          : Factor w/ 883 levels "20111225BOSNYK",..: 1 1 1 1 1 1 1 1 1
##  $ Date            : chr  "20111225" "20111225" "20111225" "20111225" ...
##  $ AwayT           : chr  "BOS" "BOS" "BOS" "BOS" ...
##  $ HomeT           : chr  "NYK" "NYK" "NYK" "NYK" ...
##  $ StartTime       : chr  "0:48:00" "0:42:59" "0:42:17" "0:41:42" ...
##  $ EndTime         : chr  "0:42:59" "0:42:17" "0:41:42" "0:40:35" ...
##  $ ElapsedTime     : chr  "0:05:01" "0:00:42" "0:00:35" "0:01:07" ...
##  $ ElapsedSecs     : int  301 42 35 67 58 35 146 30 6 165 ...
##  $ HomePlayer1ID   : int  843 843 843 990 990 990 990 990 990 843 ...
##  $ HomePlayer2ID   : int  915 915 915 915 752 752 752 25 25 990 ...
##  $ HomePlayer3ID   : int  25 25 25 25 25 25 25 315 315 315 ...
##  $ HomePlayer4ID   : int  537 537 537 315 315 315 315 597 597 537 ...
##  $ HomePlayer5ID   : int  118 118 118 118 118 988 988 988 988 597 ...
##  $ AwayPlayer1ID   : int  553 553 553 553 553 553 876 876 553 553 ...
##  $ AwayPlayer2ID   : int  166 166 166 166 419 419 419 419 876 876 ...
##  $ AwayPlayer3ID   : int  265 14 14 14 14 14 14 14 14 14 ...
##  $ AwayPlayer4ID   : int  285 83 181 181 181 181 181 181 181 181 ...
##  $ AwayPlayer5ID   : int  143 285 83 83 83 83 285 285 285 285 ...
##  $ HomePlayer1Name : chr  "Douglas" "Douglas" "Douglas" "Shumpert" ...
##  $ HomePlayer2Name : chr  "Fields" "Fields" "Fields" "Fields" ...
##  $ HomePlayer3Name : chr  "Anthony" "Anthony" "Anthony" "Anthony" ...
```

Adjusted plus/minus

PM for possessions

```
nba12$PM_Home <- nba12$PointsScoredHome/nba12$PossessionsHome
nba12$PM_Away <- nba12$PointsScoredAway/nba12$PossessionsAway
nba12$Margin<-nba12$PM_Home-nba12$PM_Away

summary(nba12$PM_Home)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00    0.50    1.00     Inf    1.62     Inf    2085
```

```
summary(nba12$PM_Away)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  0.0000  0.4444  1.0000     Inf  1.5000     Inf    2120
```

```
summary(nba12$Margin)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    -Inf -0.6667  0.0000     NaN  0.6918     Inf    3576
```

Adjusted plus/minus

We have a choice, remove cases with not number values, or impute with the mean

We will remove

```
nba6$PM<-nba12$Margin
nba6$Secs<-nba12$ElapsedSecs/60
nba7<-nba6[is.finite(nba6$PM),]
dim(nba7)
```

```
## [1] 21642   373
```

Adjusted plus/minus

The model

```
model3<-lm(PM~.,data=nba7)

coeff3<-sort(coefficients(model3), decreasing=T)
# basic cleaning
name1<-names(coeff3)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
name2$rank <- rownames(name2)
rownames(name2) <- c()
name2 <- name2 %>%
  tidyr::drop_na()
names(coeff3)<-name2$PlayerTrueName
```

Adjusted plus/minus

```
# top 10
coeff3[1:10]
```

```
##     Nash, Steve  Nelson, Jameer    Mahinmi, Ian  Hudson, Lester    Duhon, Chris
##       0.3479366       0.3374820       0.3280165       0.3243823       0.2744698
##   Millsap, Paul   Kapono, Jason Humphries, Kris    Rondo, Rajon  Curry, Stephen
##       0.2454976       0.2445955       0.2423881       0.2405524       0.2374232
```

Steve Nash, Phoenix, https://goo.gl/BczeBv

Adjusted plus/minus

Interpretations: If you take 4 average players and Steve Nash than the team will make by 0.35 points/per possession more then another team with 5 average players

```
# top 10
coeff3[1:10]
```

```
##     Nash, Steve  Nelson, Jameer    Mahinmi, Ian  Hudson, Lester    Duhon, Chris
##       0.3479366       0.3374820       0.3280165       0.3243823       0.2744698
##   Millsap, Paul   Kapono, Jason Humphries, Kris    Rondo, Rajon  Curry, Stephen
##       0.2454976       0.2445955       0.2423881       0.2405524       0.2374232
```

Adjusted plus/minus

Where is Durant ?

```
name2 %>%
  filter(PlayerTrueName == "Durant, Kevin")
```

```
##   name1 PlayerTrueName rank
## 1   714  Durant, Kevin  102
```

Where is LeBron ?

```
name2 %>%
  filter(PlayerTrueName == "James, LeBron")
```

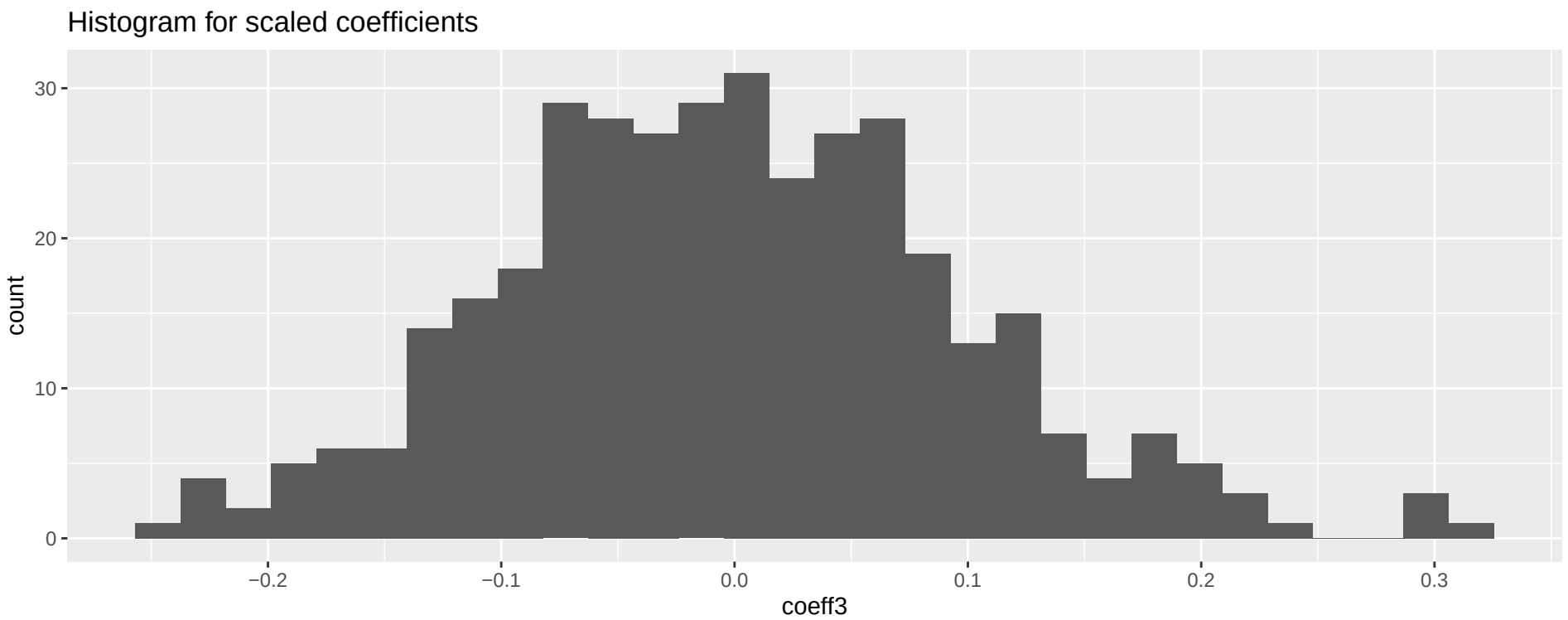
```
##   name1 PlayerTrueName rank
## 1   260  James, LeBron   20
```

Adjusted plus/minus

We can scale the Plus/Minus values around the mean, such as the positive coefficient will show that the player is doing better than the average and negative will mean that the player is doing worst than the average.

```
coeff3<-coeff3-mean(coeff3)
```

```
ggplot()+aes(coeff3)+geom_histogram()+
  ggtitle("Histogram for scaled coefficients")
```



Other considerations for regression

- Weighted by the duration of the stint, so longer time periods have bigger weights
- Weighting cases by clutch time vs garbage time

Clutch Time: During the 4th quarter or overtime, with less than five minutes remaining, and neither team ahead by more than X points.

Garbage Time: 4th quarter and overtime where either team has a lead of X points

The value of X varies for different analyses, most often 5 and 10.

Weighted regression

Weighted regression

```
nba7$Secs<-nba7$Secs/60
model4<-lm(PM~.,data=nba7, weights = Secs)
coeff4<-sort(coefficients(model4), decreasing=T)
# basic cleaning
name1<-names(coeff4)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
names(coeff4)<-name2$PlayerTrueName
coeff[1:10]
```

```
##     Benson, Keith         Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032          1.8681832          1.5567575          1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331          0.8786333          0.8486712          0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618          0.8188564
```

Adjusted plus/minus

- The APM is not an indicator on how good or bad the player is.
- The APM shows how helpful where the players for their team success.

APM regressed against other statistics

Dan Rosenbaum suggests to run the following regression to understand the impact of the box score statistics on APM

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_@ + ...\beta_k X_k + \epsilon$$

Where Y is the coefficients of the players from Plus/Minus model,

$X_1, ..X_k$ are box score statistics

APM regressed against other statistics

The results of the regression are the following:

Each coefficient shows the effect of the players box-score statistics on the APM

Basketball Statistic	Coefficient Estimate	Standard Error	Sample Mean	Standard Deviation
Pure adjusted plus/minus rating (dependent variable)	N/A		0.00	4.99
Points per 40 minutes	0.757	0.231	15.55	4.37
Effective field goal attempts per 40 minutes	-1.879	0.402	15.03	3.89
(Effective field goal attempts per 40 minutes)^2	0.026	0.010	241.1	123.3
Three point attempts per 40 minutes	0.323	0.139	2.46	2.21
Free throw attempts per 40 minutes	0.366	0.217	4.02	1.78
Assists per 40 minutes	-0.293	0.277	3.54	2.09
Offensive rebounds per 40 minutes	0.054	0.333	2.00	1.23
Defensive rebounds per 40 minutes	-0.615	0.271	4.99	1.82
Turnovers per 40 minutes	-2.171	0.480	2.36	0.65
Steals per 40 minutes	2.166	0.496	1.31	0.45
Blocks per 40 minutes	1.564	0.327	0.84	0.84
Personal fouls per 40 minutes	0.392	0.281	3.57	1.13
(Points per 40 * Assists per 40 * Rebounds per 40)$^{1/3}$	2.262	0.486	6.71	1.38
Minutes per game	0.108	0.037	27.48	8.44
Intercept	-0.043	3.083	N/A	
R^2	0.4397			
Sample size	420			

Other metrics

- Box-Score PM (Statistical Plus/Minus)- use Box-Score statistics for each stint of time of time for each player as inputs (steals, rebounds, offensive rating, etc)
- ESPN-Real Plus/Minus – is calculated by ESPN again using some sort of regression analysis.
- Offensive Plus/Minus and Defensive Plus/Minus
- Win shares: Contribution of each player into wins of the team.
- VORP – Value Over Replacement Level

Where to find the ratings?

Plus/Minus

- Best lineup by plus/minus
  https://stats.nba.com/lineups/traditional/
- ESPN: Offensive Real Plus/Minus, Defensive real Plus/Minus, RPM and Wins share (from 2014) https://www.espn.com/nba/statistics/rpm/_/sort/RPM
- Plus minus by player
  https://www.basketball-reference.com/players/d/duranke01/gamelog/2012/

Other Metrics

- WINVAL by Wayne Winston: The exact method of calculating tha Adjusted P/M is not public, however.

Margin of victory = sum(Rating of all 5 home team players) - sum(Rating of all 5 away team players)